A dynamical perspective on QEC

Using ZX/W rewrites to go beyond depolarizing noise

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Outline

test

Motivation

Motivation

- QEC codes described in the stabilizer formalism not directly implementable, either due to the **weight** of operators or the required **connectivity** on qubits
- Real-world implementations require intermediary steps in the measurement of stabilizers

  $\Rightarrow$ **Syndrome extraction**

Motivation

- Modeling Errorchannels, detectors, corrections in a diagramatic representation can expose symmetries as patterns for simplification routines

  $\Rightarrow$ We could feed our problems into a compiler
- In many ways, both hardware tailoring and QEC itself are just compilation problems

ZX and rewrites

ZX is a monoidal category whose objects are spiders and whose morphisms are called "rewrite rules"

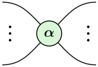
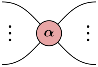
Z-spiders	**X-spiders**
 $= \lvert 0\cdots0\rangle\langle 0\cdots0\rvert + e^{i\alpha}\lvert 1\cdots1\rangle\langle 1\cdots1\rvert$	 $= \lvert +\cdots+\rangle\langle +\cdots+\rvert + e^{i\alpha}\lvert -\cdots-\rangle\langle -\cdots-\rvert$

Figure: (unnormalized) Z and X spider definitions

Core ZX rules

- Equality of any two diagrams D_1 and D_2 can be proven using a generating set of local rewrite identities.

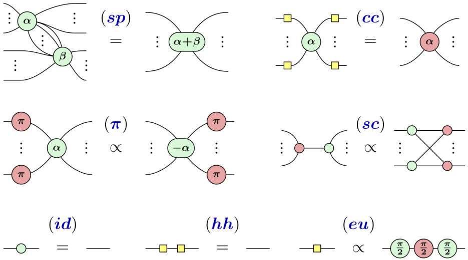


Figure: Generating set of 7 local rewrite identities

Teleportation

Teleportation protocol

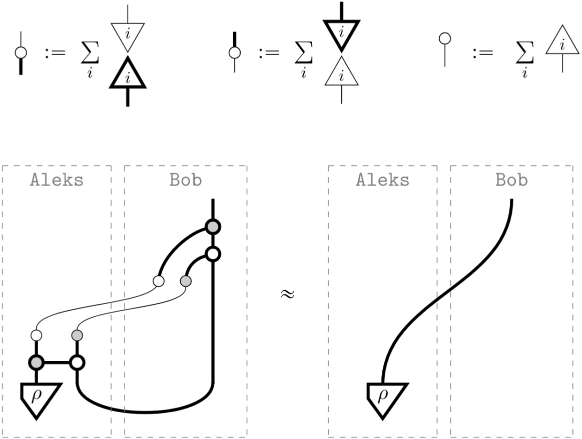


Figure: In the teleportation protocol, the classical in/outputs are δ-functions, which is why sometimes they are just denoted with a and b labels

Pauliwebs

- Pauliwebs are syntactic sugar to ZX diagrams, used to track local gauge symmetries on a diagram
- For the Clifford fragment of ZX, i.e. diagrams whose spiders only have phases in integer multiples of $\frac{\pi}{2}$, two conditions must hold locally for every spider:
  - a spider with $k\pi$ phase can have
    1. An even number of legs marked in its own color and/or
    2. All or none of its legs marked in opposite color
  - a spider with $\pm\frac{\pi}{2}$ phase can have
    1. An even number of legs marked in its own color and none in the opposite color
    2. An odd number in its own color and all legs in the opposite color

Pauliwebs: Examples

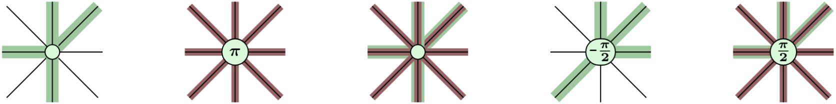


Figure: Examples of valid Pauliwebs. Each of these is semantically equivalent to the underlying spider without any webs

Pauliwebs: Examples

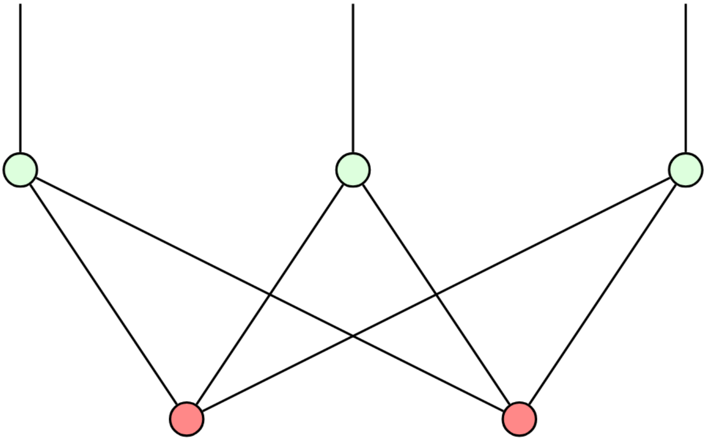


Figure: Using pauliweb firings, we can read off the stabilizers of this stabilizer state.

The QEC perspective

Detectors

- A detector is a set of measurements outcomes whose joint parity is promised to be even in the absence of a detection event.
- Simplest case: Measuring a stabilizer element twice
  - If $S_i \in S$ then $(M_{S_{i,1}} + M_{S_{i,2}})\text{mod}2 = 0$ in the absence of the detectable error E_i for that detector
  - This follows from $\forall S_i, S_j \in S : [S_i, S_j] = 0$ and $\forall E_i \in E_{detectable} : \exists S_i \in S : \{E_i, S_i\} = 0$
- But: more complex detector configurations involving 3 or more measurements possible

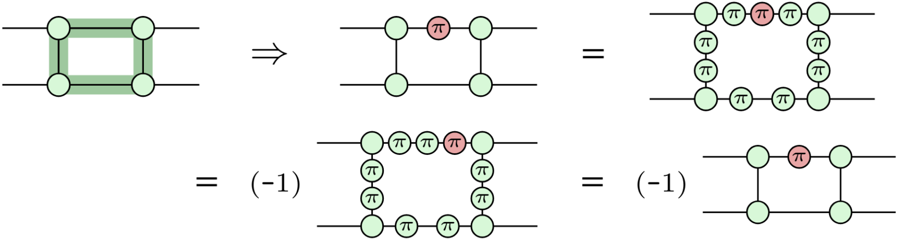


Figure: A diagram D that can be rewritten to -D is equal to 0.

$\Rightarrow$ The map described by the diagram cannot occur.

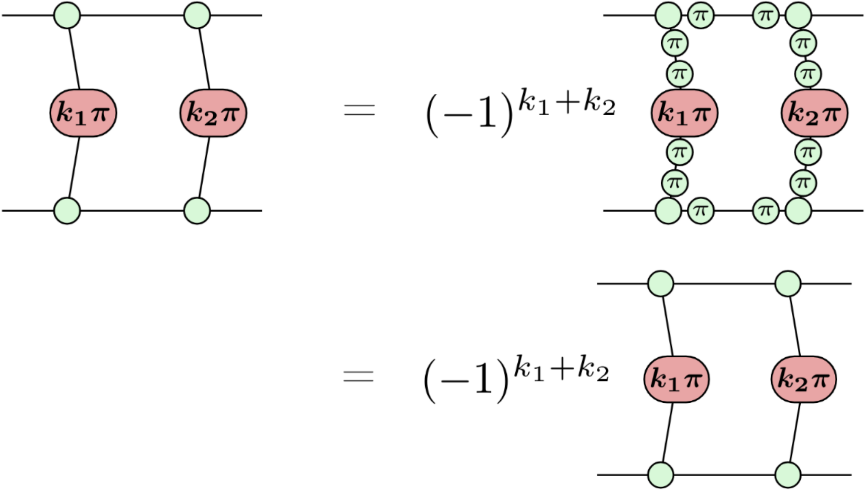


Figure: Case with general measurement results k_1 and k_2.

$\Rightarrow$ The map described by the diagram cannot occur.

Distance preservation

- ZX distance: smallest nontrivial error $\mathcal{E}$ s.t. $[\![D + \mathcal{E}]\!] \neq 0$.
- A **distance-preserving** rewrite, is a rewrite that preserves ZX distance.

Distance preservation

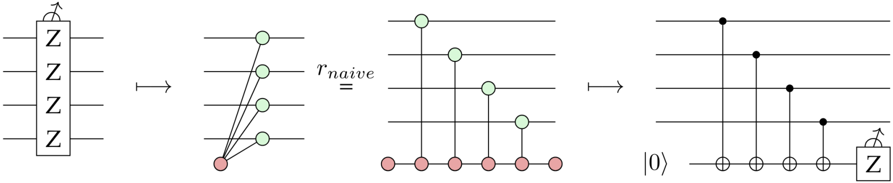


Figure: example of a naive non-distance-preserving rewrite/ circuit extraction

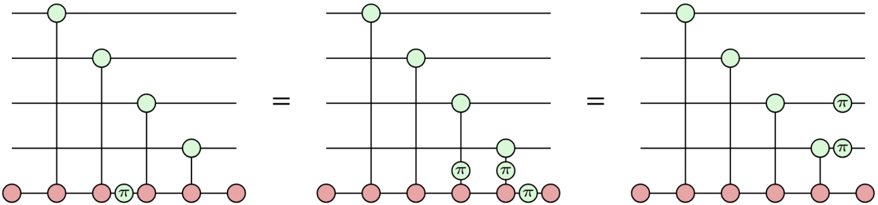


Figure: Errors propagate onto multiple qubits under this naive rewrite

Distance preservation

These two rewrites are enough to distance-preservingly rewrite arbitrary-legged spiders:

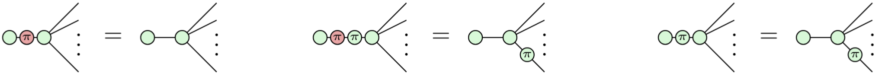


Figure: Unfusing a dangling leg is distance-preserving

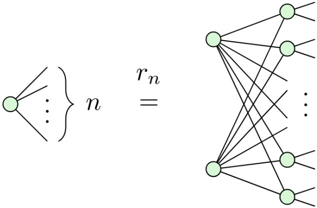


Figure: Distance-preserving rewrite of a spider with n legs where n mod 2 = 1

Detectors: ZX example

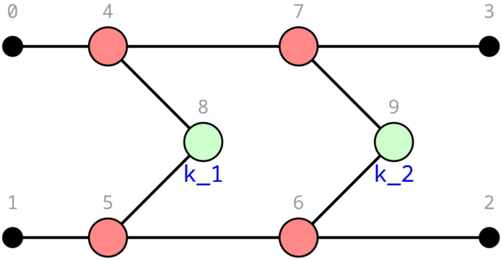


Figure: Two consecutive XX measurements. The parity of $k_1 + k_2$ will be even in the absence of errors.

Detectors: ZX example

- A ZX diagram representing the Choi state of two consecutive XX measurements, brought into bipartite form

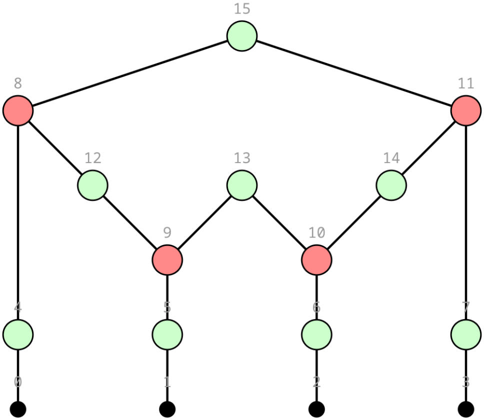


Figure: two XX measurements Choi state (k_1 and k_2 on 12 and 14)

Firing verification matrix

Bringing the graph into bipartite form, the conditions for phaseless diagrams now simplify to:

1. Every internal spider must have an even number of neighboring spiders firing
2. Output spiders cannot fire

A firing vector is therefore **valid** if for all internal spiders q it satisfies

$$\sum_{x:(q,x)\in E} \mathbf{x} = 0 \bmod 2 \tag{1}$$

We can encode this in a **Firing verification matrix**, whereby we want our firing vectors to satisfy $M_D \vec{v} = 0$ with:

$$M_D = \left(\begin{array}{c|c} I_n & \multirow{2}{*}{N} \\ \emptyset & \end{array} \right) \tag{2}$$

Firing verification matrix (non-CSS)

For more general non-CSS stabilizer codes, $\pm\frac{\pi}{2}$ spiders are needed in their diagrams.
In order to account for $\pm i$ phases accrued by firing, internal spiders h with phase $\pm\frac{\pi}{2}$ must then additionally satisfy:

$$\sum_{x:(h,x)\in E} \mathbf{x} = \mathrm{h} \bmod 2 \tag{3}$$

giving us an updated verification matrix

$$M_D = \left(\begin{array}{c|c} I_n & \mathrm{N} \\ \emptyset & \end{array}\right) - \left(\begin{array}{c|c} \emptyset & \emptyset \\ \emptyset & I_r \end{array}\right) \tag{4}$$

where r is the number of internal $\pm\frac{\pi}{2}$ spiders.

Computing Detection Webs on a CSS code

- Fire spiders in their opposite color according to **valid firing vector**

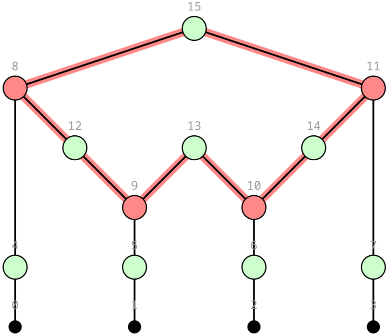


Figure: Fired spiders 8, 9, 10 and 11

Some rewrites and their detection webs

Example: Dist-pres rewritten XXX measurement

- Detection webs give a more circuit-level understanding of noise

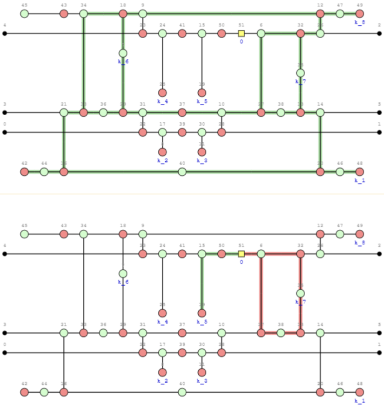


Figure: Rewritten XXX measurements

Example: Steane Code

Theoretically, the pure stabilizer measurements of this code involve weight-4 measurements:

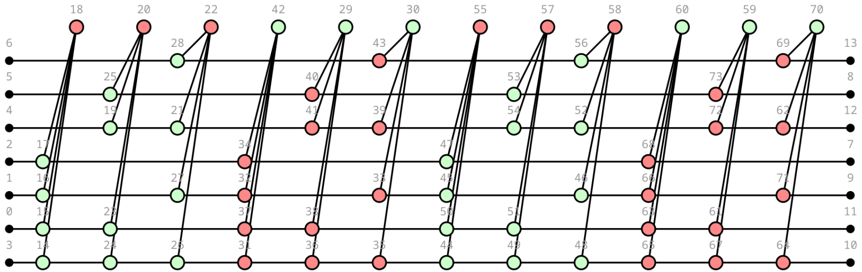


Figure: two rounds of the Steane code stabilizer measurements

Example: Steane Code

Theoretically, the pure stabilizer measurements of this code involve weight-4 measurements:

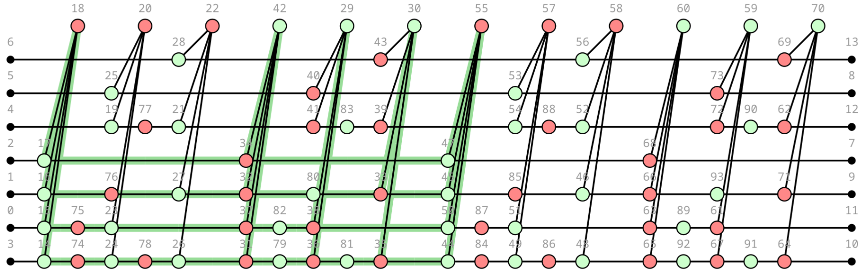
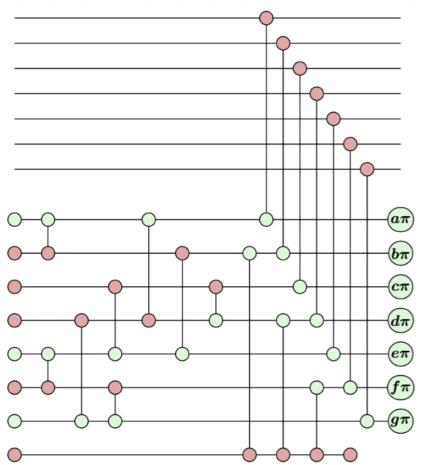


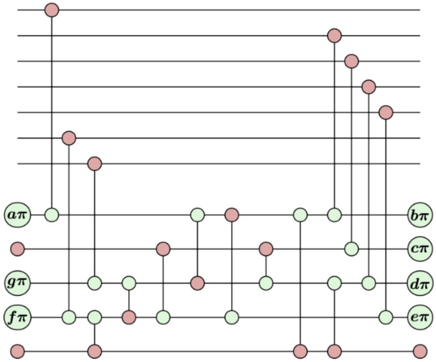
Figure: A detector on the theoretical steane code

Example: Steane Code

We can use distance preserving rewrites to get more efficient realisations of previously known fault-tolerant syndrome extraction techniques



(a) Steane-Style syndrome extraction



(b) Rewrite using fewer ancillas and CNOTs but preserving distance

Detection webs on rewrite of steane-style Steane measurements

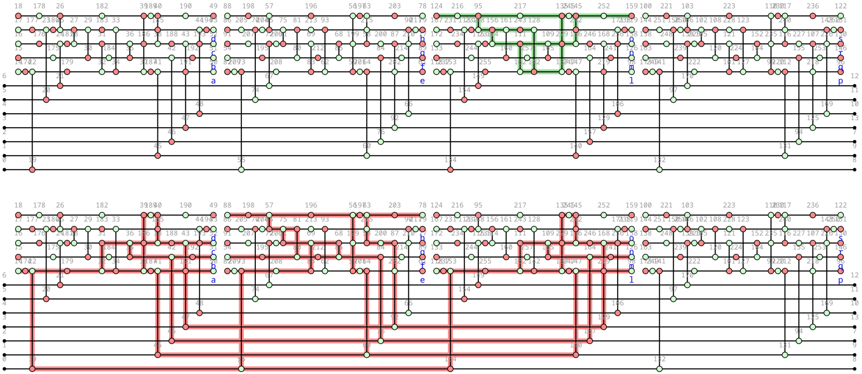


Figure: Rewritten Steane code and two of its detection regions

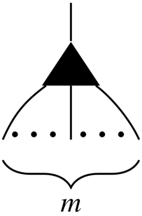
Outlook

More general noise models

Error Channels in ZX

- Since an error channel represents an entanglement to the outside world, it is necessarily non-unitary

  $\Rightarrow$ To represent a nontrivial error channel in ZX we need a good way to represent sums of diagrams
- The W spider is a useful adition to represent these concisely



$$= \underbrace{|0\cdots0\rangle}_{m}\langle 0| + \sum_{k=1}^{m} \overbrace{|\underbrace{0\cdots0}_{k-1}10\cdots0\rangle}^{m}\langle 1|$$

Figure: Definition of W spider

As a consequence, sums of diagrams become very natural with:

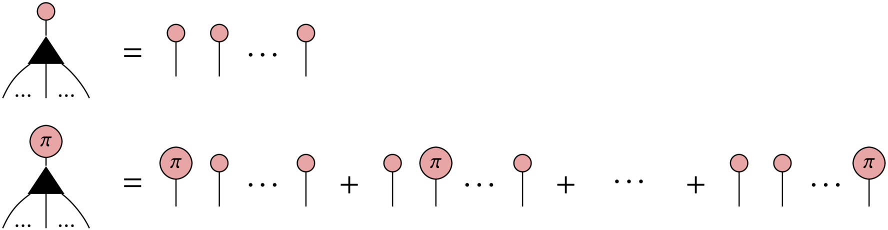
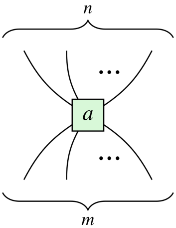


Figure: Effect of W spider with 0 and π gadget on diagram.

ZXW


$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

(a) m=2 W spider



$$= |0\rangle^{\otimes m} \langle 0|^{\otimes n} + a\,|1\rangle^{\otimes m} \langle 1|^{\otimes n}$$

(b) Box Spider

ZXW for general hamiltonians

Then to represent a hamiltonian of commuting paulis $\mathcal{H} = \sum_{i=1}^{n} \alpha_i \bigotimes_{j=1}^{m} P_{ij}$

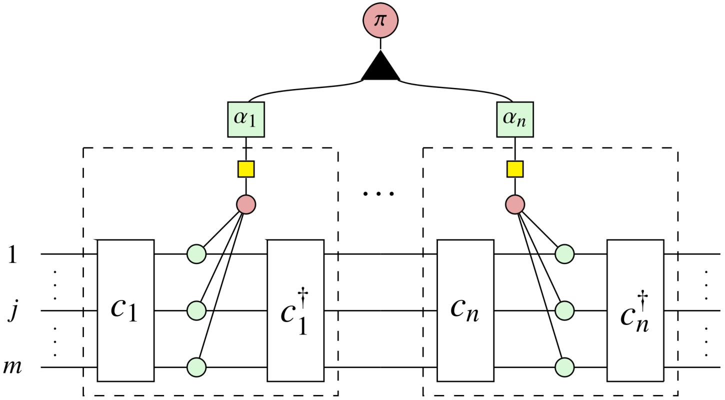
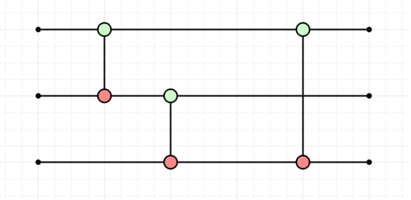


Figure: One way to represent a general hamiltonian in ZX. Here c_{ij} are the clifford conjugations of Z to P_{ij}

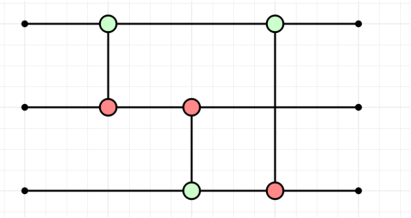
A path to computing thresholds

Fidelity in ZX

Hilbert-Schmidt fidelity in ZX



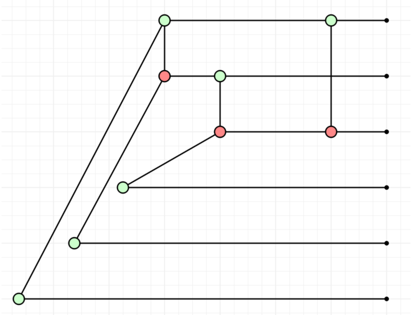
(a) c1



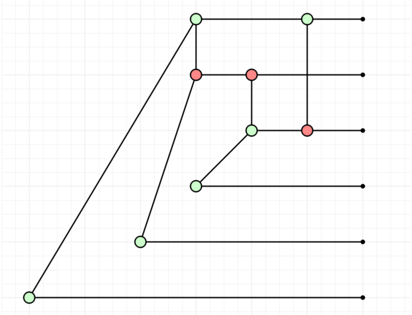
(b) c2

Figure: Two diagrams c1 and c2

Hilbert-Schmidt fidelity in ZX



(a) choi1

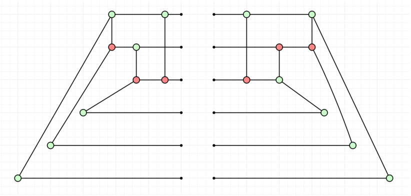


(b) choi2

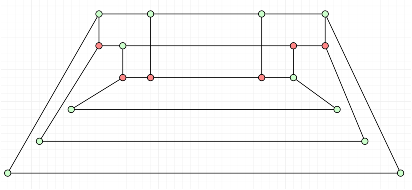
Figure: Choi states of diagrams c1 and c2 respectively

HS Fidelity

Now taking the adjoint of the choi state corresponding to c2 and putting the output of choi1 as input, we obtain a closed diagram, i.e. a number:



(a) c1 choi state with adjoint c2 choi state next to it



(b) composition of choi1 and adjoint of choi2

Figure: Since this is a closed diagram, it can be simplified to the scalar value $\frac{2}{9}$

HS Fidelity

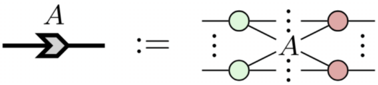
We can therefore use ZX simplification routines, which reflect symmetries in the map, to **more efficiently compile fidelity** values. The previous example corresponds to the **Hilbert-Schmidt Fidelity**

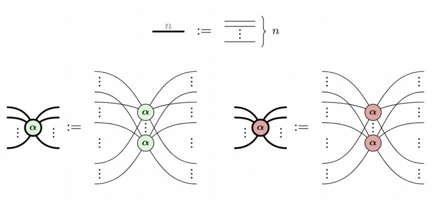
$$\mathbb{F}_{HS} = \frac{Tr\left(\mathbb{J}(A)\mathbb{J}(B)^{\dagger}\right)}{||\mathbb{J}(A)||_{HS}} \tag{5}$$

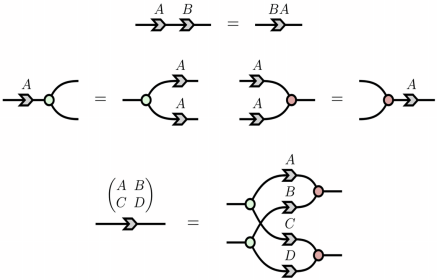
scalable ZX for reasoning about scalable codes

Scalable ZX for CSS codes

This grey arrow together with these rules for red and green spiders on thick lines, gives us the scalable zx





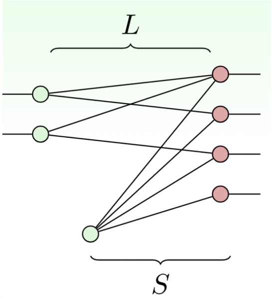


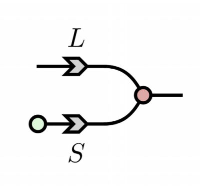
Scalable ZX for CSS codes

So coming back to the 4,2,2 code, the normal form of its encoding is:

This gives us logical operators XXII, XIXI and stabilizer XXXX

Similarly for the Z stabilizer





$$L := \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \qquad S := \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

"Physics is all about Symmetries."

– Daniel Reich, Advanced Quantum Mechanics